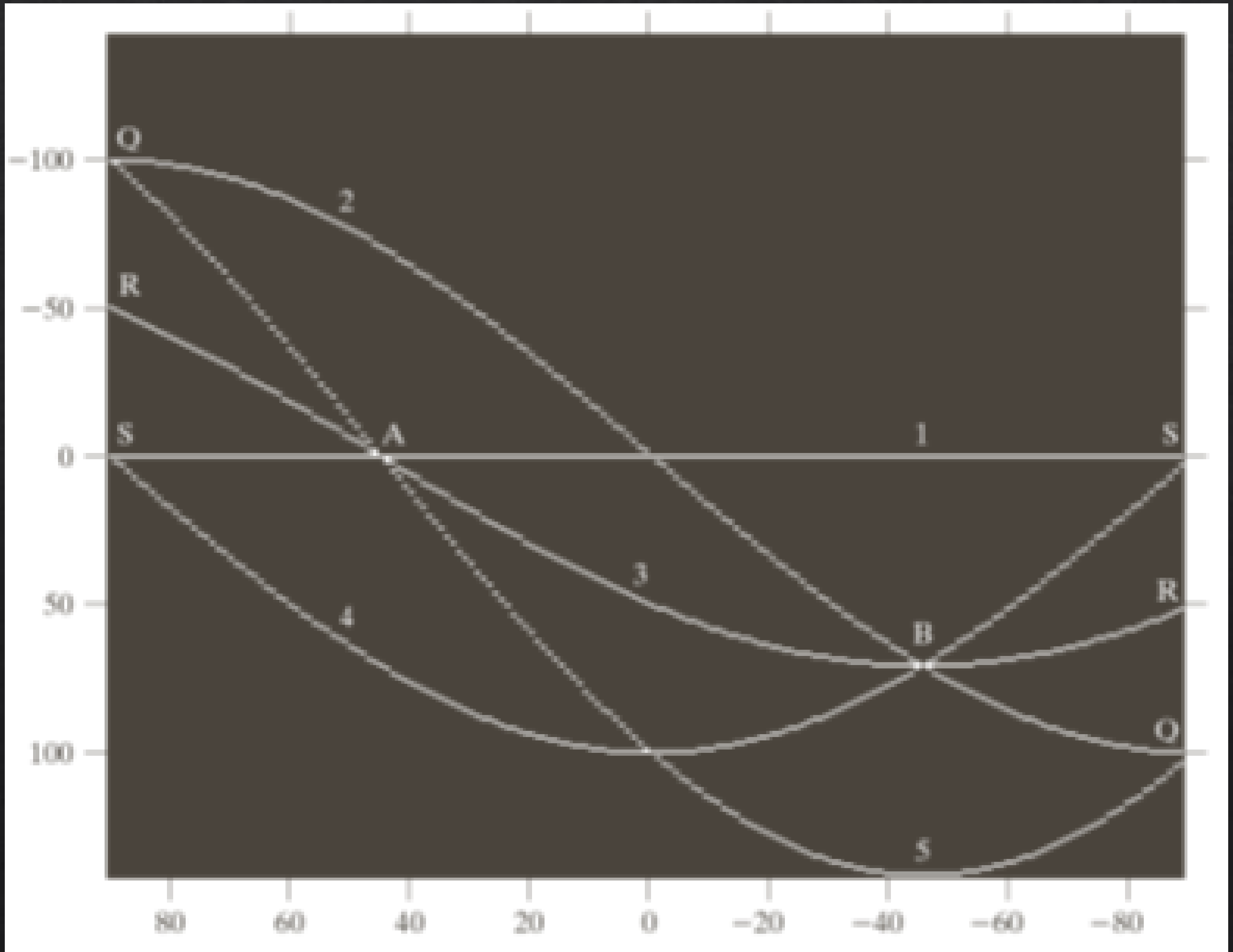
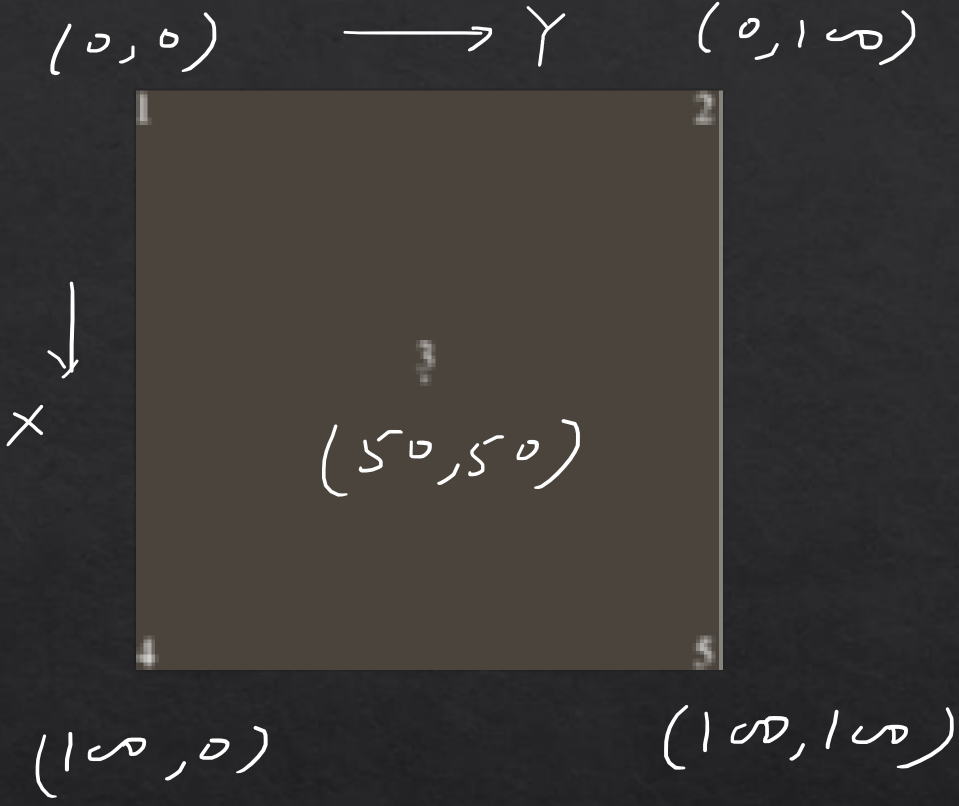
Hough Transform

Line detection

Example

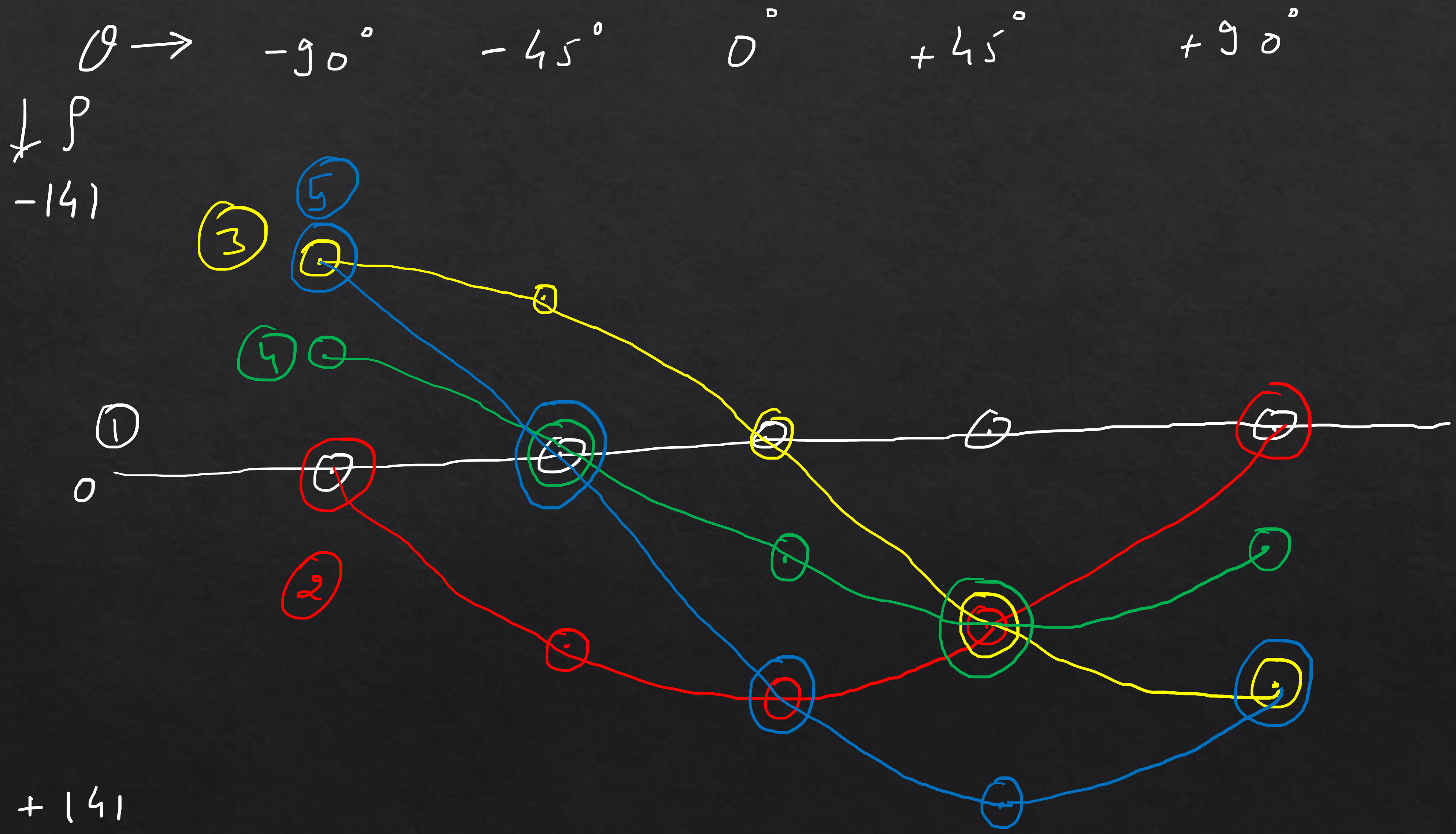




$X \cos\theta + Y \sin\theta = \rho$

P.L. ↓ \ θ →	-90°	-45°	0°	$+45^\circ$	$+90^\circ$
(0, 0)	0	0	0	0	0
(100, 0)	0	71	100	71	0
(0, 100)	−100	−71	0	71	100
(50, 50)	−50	0	50	71	50
(100, 100)	−100	0	100	141	100

$$x\cos\theta + y\sin\theta = \rho$$



ρ	θ	Count
0	+90	+1
0	−90	+1
0	−45	+2
0	0	+1
71	+45	+2
100	0	+1
100	+90	+1
−100	−90	+1

All remaining cells $\rightarrow$ 0

$$x\cos\theta + y\sin\theta = \rho$$

① $(0, 0) \Rightarrow \rho = 0$

$\rho = 0$ & $\theta = \pm 90^\circ$ $+1$

② $(100, 0) \Rightarrow 100\cos\theta = \rho$

$\rho = 0$ & $\theta = 0^\circ$ $+1$

③ $(0, 100) \Rightarrow 100\sin\theta = \rho$

$\rho = 71$ & $\theta = +45^\circ$ $+1$

④ $(50, 50) \Rightarrow 50(\cos\theta + \sin\theta) = \rho$

$\rho = 0$ & $\theta = -45^\circ$ $+1$

$\rho = 71$ & $\theta = +45^\circ$ $+1$

⑤ $(100, 100) \Rightarrow 100(\cos\theta + \sin\theta) = \rho$

$\rho = -100$ & $\theta = -90^\circ$ $+1$

$\rho = 100$ & $\theta = 0^\circ$ $+1$

$\rho = 0$ & $\theta = -45^\circ$ $+1$

$\rho = 100$ & $\theta = +90^\circ$ $+1$

$$x\cos\theta + y\sin\theta = \rho$$

$\rho = 0$ & $\theta = -45°$

Eqn of first line

$$x - y = 0 \quad (1)$$

$\rho = 71$ & $\theta = +45°$

Eqn of second line

$$x + y = 100 \quad (2)$$